

Handiko Gesang Anugrah Sejati

RETAIL ALGORITHMIC ANALYST / TRADER

📍 Jakarta, Indonesia

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🌐 handiko.github.io/algo

PROFESSIONAL SUMMARY

Quantitative-focused Algorithmic Trader and Developer with a deep understanding of market microstructure across Crypto, Forex, and Equities. Specialized in developing automated execution systems in MQL5, with a strong emphasis on liquidity analysis, risk mitigation, and order book dynamics. Seeking to transition technical expertise in market mechanics into Crypto Business Operations to optimize exchange liquidity and execution quality.

TECHNICAL SKILLS

Market Operation: Order Book Management, Slippage Analysis, Market Maker Performance Tracking, Liquidity Provision

Market Analysis: Order Book Dynamics, Liquidity Mechanics, Bid-Ask Spread Optimization, Hedging

Risk Management: Inventory Risk Modelling (Avellaneda-Stoikov model), Anomaly Detection

Trading Systems: MQL5 (Expert Advisor), MetaTrader 5, Algorithmic Strategy Design

Data & Tools: MQL5 Backtesting & Optimization, Risk Management Modules (Python), Financial Data Analysis (Python), XGBoost

Core Competencies: P&L Tracking, Market Anomaly Detection, Quantitative Research

RELEVANT EXPERIENCE

Independent Algorithmic Trader & Developer

2023 – Present

- Developed and deployed a live market maker monitoring and simulation dashboard based on the Avellaneda-Stoikov framework.
- Engineered and deployed custom automated trading systems and Expert Advisors (EAs) in MQL5 for Forex and Crypto markets.
- Analyzed market microstructure parameters to refine entry/exit logic, aiming at minimizing execution slippage and maximizing filled orders within low-liquidity environments.
- Managed a multi-asset diversified portfolio across Cryptocurrencies, Forex, and Equities with structural P&L metrics tracking and attribution.
- Conducted extensive backtesting and multi-variable parameter optimization suites to isolate market inefficiencies and anomalies.

SELECTED PROJECTS

Machine Learning Stock Backtesting Framework

Can a machine learning model beat buy-and-hold on Indonesian stocks? This project builds, tests, and rigorously evaluates a trading strategy powered by XGBoost — one of the most battle-tested ML algorithms in quantitative finance.

🔗 **Live Project:** handiko.github.io/machine-learning-stock-backtesting

Avellaneda-Stoikov Market Making Simulation

The simulator models a market maker (MM) who provides liquidity by simultaneously posting bid and ask quotes. The MM's goal is to capture the spread while managing Inventory Risk (the risk of price movement against a held position) and Adverse Selection (the risk of trading against informed participants).

🔗 **Live Project:** handiko.github.io/market-maker-sim

USDT-IDR Market Monitor

A high-fidelity monitoring terminal for the USDT/IDR pair, built for liquidity analysis and execution simulation across Indonesian exchanges. This project focuses on minimizing latency and maximizing the visibility of order book depth.

 **Live Project:** handiko.github.io/USDT-IDR-Monitoring-Dashboard

MOST RECENT WORK HISTORY

PT. Imani Prima — Firmware Engineer

2020 – 2025

- Managed research and development (R&D) for AIS (Automatic Identification System) transmitters and receivers deployed in national maritime and fisheries institution infrastructure projects.
- Applied rigorous attention to detail and data-driven decisions to resolve engineering roadblocks and collaborate across functional product teams to improve system testing workflows.

EDUCATION

Universitas Gadjah Mada (UGM)

Graduated

Bachelor of Engineering in Engineering Physics

- **Specialization:** Instrumentation and Control.
- **Final Project:** Multi-channel modulators for Software Defined Radio systems.